

DAY 5 — DATASET RESEARCH

Name: Praneetha AS (aspraneetha29@gmail.com)

Role: AI/ML Intern

Date: 15/05/2026

Project Folder: Raritone

Platforms explored: Kaggle, google colab, GitHub

1. Introduction

Datasets play a crucial role in machine learning and artificial intelligence. They are used to train, validate, and test models so that systems can learn patterns and make predictions.

In this task, a fashion-related dataset was explored using Kaggle to understand its structure, components, and applications in AI-based systems.

2. Platforms Used

- **Kaggle** — Used for exploring and understanding datasets
- **Google Colab** — Can be used for executing dataset-related code (optional)

3. Dataset Selected

The dataset selected for this task is:

Fashion-MNIST Dataset

4. Dataset Overview

Fashion-MNIST is a dataset of clothing images used for training machine learning models. It consists of grayscale images of various fashion items and is widely used for classification tasks.

Each image represents a specific clothing category and is labeled accordingly.

5. Dataset Structure

The dataset is divided into:

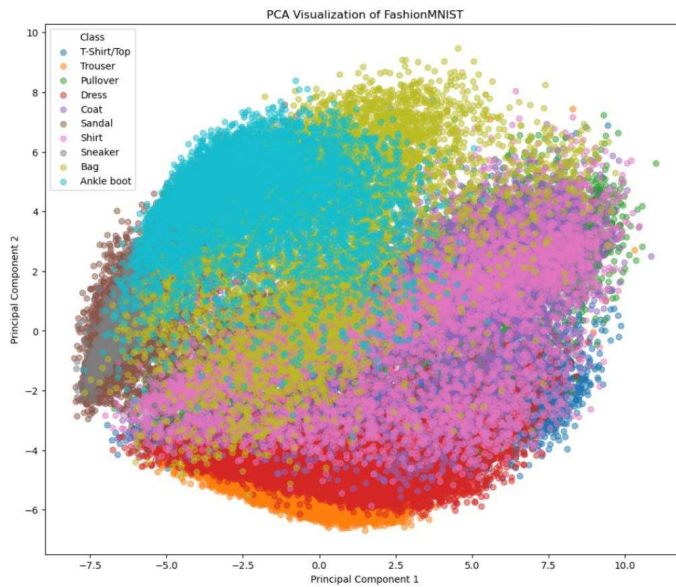
- Training dataset
- Testing dataset
- Labels corresponding to each image

Each image has the following properties:

- Size: 28×28 pixels
 - Format: Grayscale
 - Type: Image data
-

6. Sample Dataset Images





6.1 Google Colab Implementation

Google Colab is an online Python environment that allows users to run machine learning code without installing any software locally. It provides access to libraries such as TensorFlow and is useful for experimenting with datasets.

In this task, a simple code snippet was used in Google Colab to load and explore the Fashion-MNIST dataset.

Sample Code

```
from tensorflow.keras.datasets import fashion_mnist

(trainX, trainY), (testX, testY) = fashion_mnist.load_data()

print("Training Data Shape:", trainX.shape)
print("Testing Data Shape:", testX.shape)
```

Explanation

- The dataset is loaded using TensorFlow
- Training and testing datasets are separated
- The shape of the data shows the number of images and their dimensions

Observation

- The dataset contains thousands of images
- Each image is of size 28×28 pixels
- The data is ready for machine learning tasks

7. Categories in Dataset

The dataset contains the following categories:

- T-shirt
- Trouser
- Pullover
- Dress
- Coat
- Sandal
- Shirt
- Sneaker
- Bag
- Ankle boot

Each category is assigned a numerical label which helps in classification tasks.

8. Observations

- The dataset is well-structured and organized
 - Images are simple and easy to process
 - Labels are clearly defined
 - Suitable for beginners in AI/ML
 - Helps in understanding image classification
-

9. Applications

The Fashion-MNIST dataset can be used in:

- Clothing classification systems
- Fashion recommendation systems
- Virtual try-on applications
- AI-based shopping platforms

10. Advantages

- Easy to use and lightweight
- Well-labeled data
- Suitable for learning and experimentation
- Widely used in AI research

11. Limitations

- Low image resolution
- Does not represent real-world complexity
- Limited variation in clothing styles

12. Conclusion

The Fashion-MNIST dataset provides a strong foundation for understanding how machine learning models work with image data. It is simple, well-structured, and suitable for beginners. This dataset can be effectively used to build basic AI models for classification and fashion-related applications.